INCH-POUND

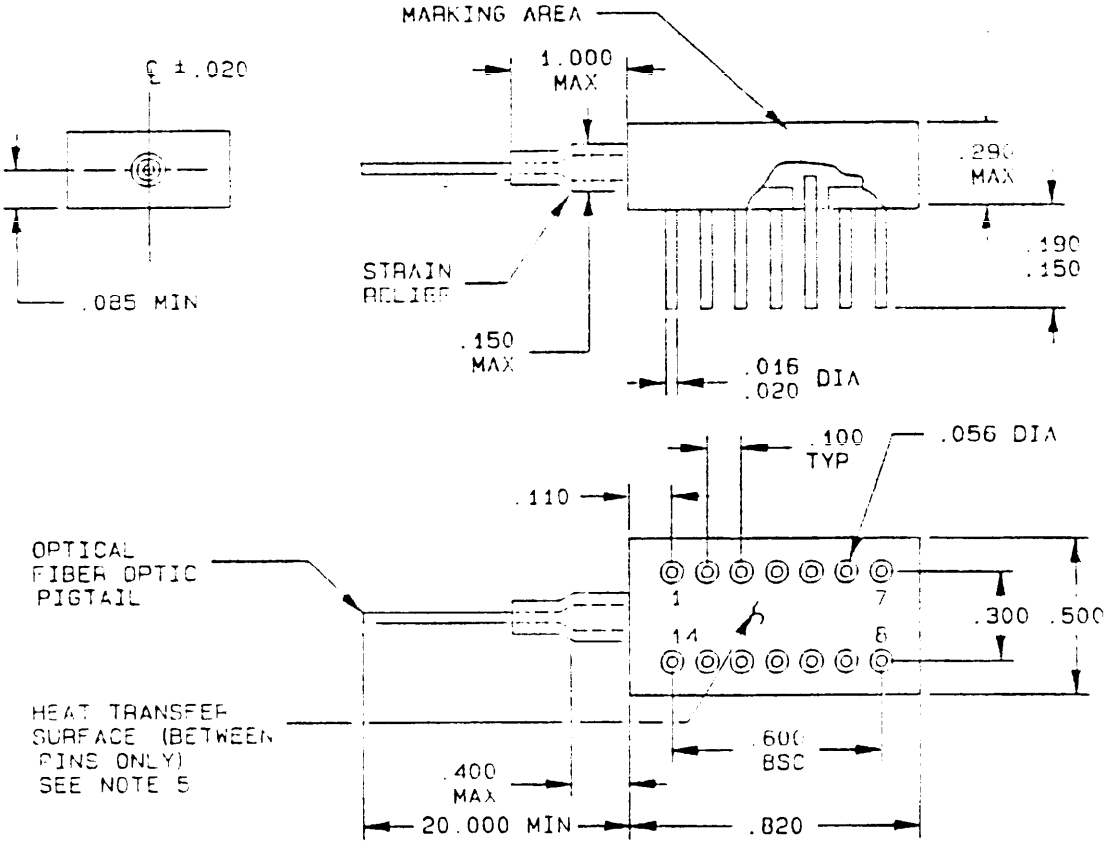
MIL-S-24622/3A
30 June 1989
SUPERSEDING
MIL-S-24622/3(CR)
23 January 1987

MILITARY SPECIFICATION SHEET

SOURCE, LED, FIBER OPTIC, 1290 NANOMETERS, GLASS PIGTAILED, HERMETICALLY SEALED, DUAL IN-LINE PACKAGE (DIP)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24622.



Inches	mm
.016	0.40
.020	0.50
.056	1.42
.085	2.16
.100	2.54
.110	2.79
.150	3.81
.190	4.82
.290	7.36
.300	7.62
.400	10.16
.500	12.70
.600	15.24
.820	20.82
1.000	25.40
20.000	508.00

FIGURE 1. LED source - (DIP) pigtail package outline and pin connections.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Pin connections

Pin number	Function
1	NC
2	NC
3	NC
4	NC
5	NC
6	Anode
7	NC
8	NC
9	Cathode
10	Ground
11	NC
12	NC
13	NC
14	NC

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Leads shall be gold (50 microinches minimum) in accordance with MIL-G-45204, type II, grade A, class 1 over nickel (100 microinches minimum) in accordance with QQ-N-290, class 1, grade C or MIL-C-26074.
5. The heat transfer surface (heatsink) shall provide the major heat transfer mechanism to remove heat from the device. Convection and radiation to the environment surrounding the device must not provide significant heat transfer for this device when it is being tested to meet this specification.
6. A yellow band with the phrase "fiber optic" included within it shall be provided on the strain relief of the fiber optic pigtail.

FIGURE 1. LED source - (DIP) pigtail package outline and pin connections - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1.

Pigtail fibers: As specified by DOD-F-49291/1 with the exception that the buffer coating be upjacketed to a diameter of 900 µm ±50 µm. The fiber end termination shall be unconnectorized.

Fiber retention:

Tension: Fiber pigtails shall withstand a minimum tensile load of 5 newtons for 30 seconds, minimum.

Electrical and optical requirements: See table I.

Environmental requirements:

Temperature range:

Operating: -46°C to +85°C.

Storage: -57°C to +85°C.

Nuclear radiation resistance: The nuclear radiation resistance characteristics of this light emitting diode are classified and are contained in an appendix to this specification. Application to receive this appendix must be made through the US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ, 07703-5016. Information concerning security clearance classification and "need to know" must be detailed in the request. When measured 15 minutes after the specified irradiation, the light emitting diode device shall satisfy all the requirements contained herein, except that the peak power output minimums are reduced to 87 percent (-0.6 dB) at 150 mA peak.

Screening: Screening shall be conducted on 100 percent of all devices prior to groups A, B, and C inspections and shall be in accordance with methods specified in table II. Tests shall be performed in accordance with method 5008 of MIL-STD-883.

Group A inspection: Group A inspection shall consist of examinations and tests specified in table III.

Group B inspection: Group B inspection shall consist of examinations and tests specified in table IV.

Group C inspection: Group C inspection shall consist of examinations and tests specified in table V.

Part or Identifying Numbers (PIN):

M24622/3-001
M24622/3-002

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TABLE I. Electrical and optical requirements. 1/

Source dash number	Drive current I_d (mA) (peak) 2/	Maximum drive current I_{DMAX} (mA) 3/	Forward voltage V_F (V) 4/	Reverse voltage V_R (V) 5/	Central wavelength λc (nm) 6/	Wavelength drift λd (nm/°C) max	Spectral bandwidth $\Delta\lambda$ (nm) max 7/	Peak power amplitude 2/			Pulse times t_r/t_f (ns) max 8/	Linearity percent 9/	Contrast ratio (percent)
								Heat sink temperature	Output power (μW) Minimum	Output power (μW) Maximum			
-001	150	200	1.8 max	1.0	1290 ±20	0.7	120	-46°C	60	350	8	15 max of total ac optical output power	1.5
								+25°C	30	150			
								+85°C	18	100			
-002	150	200	1.8 max	1.0	1290 ±20	0.7	120	-46°C	60	350	8	Not applicable	Not applicable
								+25°C	30	150			
								+85°C	18	100			

1/ Unless otherwise specified, the ambient temperature shall be +25°C ±2°C.

2/ 150 mA peak drive current, 50 percent duty cycle, 10 MHz.

3/ Device to be driven at the maximum drive current for a minimum of 180 seconds.

4/ 150 mA dc drive current.

5/ The reverse current shall not exceed 30 μA at 1 volt reverse bias.

6/ λc is the wavelength at the central output amplitude (midpoint of the 50 percent power points).

7/ $\Delta\lambda$ is measured between the 50 percent power points (FWHM) and shall remain less than the value specified over the entire operating temperature range.

8/ t_r/t_f values are taken at the 10 to 90 percent amplitude points. The drive current is to be varied over 2.5 to 100 percent of 150 mA for -001 devices and 0 to 100 percent of 150 mA for -002 devices. The drive current rise and fall times shall be less than or equal to 5 ns. There shall be less than ten percent overshoot on the drive current waveform.

9/ Sinusoidal modulation at 10 MHz shall be applied to the device. The drive current shall be varied from 3.75 mA to 150 mA by this modulation. All harmonics of the fundamental 10 MHz input frequency shall be less than or equal to the value specified.

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TABLE II. Screening inspection.

Screen inspection (100%)	MIL-STD-883 method	Details
Internal visual inspection	2017	Ambient
Stabilization bake	1008	24 hours minimum at T = +85°C
Thermal shock (temperature cycling)	1010	Test condition A, steps 1 and 3 shall be 2 hours minimum
Constant acceleration	2001	Y1 orientation only at 5,000 G minimum
Hermetic seal Fine leak rate Gross leak rate	1014	 Test condition A_1, 5 x 10^{-7} atm cm^3 maximum leak rate. Test condition C_1, except that leak indicator fluid shall be maintained at +100°C ±5°C
Steady-state burn-in	1015	Condition B for 160 hours minimum. I_D = 150 mA, temperature (heatsink) = +85°C
External visual and mechanical inspection	2009	Ambient
End points: Subgroup 1 Group A		 $\Delta V_f \leq 25\%$ $\Delta P_o \leq 20\%$

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TABLE III. Group A inspection.

Inspection	Details	LTPD
Subgroup 1		
Peak output power	T = +25°C	5
Central wavelength	I_d = 150 mA peak	5
Forward voltage drop, dc		10
Subgroup 2		
Linearity 1/	T = -46°C, +25°C, and +85°C	10
Contrast ratio 1/ 2/		10
Temperature dependent emission spectra		10
Temperature dependent peak power		10
Maximum drive current, dc	Time = 130 seconds	10
Reverse bias condition	V_R = 1.0 at 30 µA max	10
Pulse rise and fall time		10

1/ Applicable to M24622/3-001 only.

2/ The contrast ratio shall be determined during the temperature dependent peak power test at the temperatures of -46°C, +25°C, and +85°C using the following equation:

$$\text{Contrast ratio (\%)} = \frac{\text{Optical power at 4 mA (µW)}}{\text{Optical power at 150 mA (µW)}} \times 100$$

Any unit with a contrast ratio greater than 1.5% at any of the specified temperatures shall be considered to have failed.

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TABLE IV. Group B inspection.

Inspection	Method	MIL-STD-883 details	LTPD
Subgroup 1 1/			
Solderability 2/	2003		15
Lead integrity	2004	Test condition C_1	15
Resistance to solvents	2015		15
Subgroup 2 1/			
Decap internal visual and mechanical	2014		15
Bond strength	2011		10
Subgroup 3			
Physical dimensions	2016	See figure 1	20
Fiber retention			10
Hermetic seal	1014	Test condition A_1 for fine leaks (leak testing 2 hours after pressurization is acceptable); test condition C_1 for gross leaks, except that leak indicator fluid shall be maintained at +100°C ±5°C.	10
End points: Subgroup 1 Group A			

1/ Failed devices from the same inspection lot may be used for all subgroups when electrical or optical end-point measurements are not required.

2/ The LTPD for solderability test applies to the number of leads inspected except in no case shall less than three devices be used to provide the number of leads required.

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TABLE V. Group C inspection.

Inspection	MIL-STD-883 method	Details	LTPD
Subgroup 1			
Thermal shock (temperature cycling)	1010	Test condition A-1, except the minimum time for steps 1 and 3 shall be 2 hours.	10
Moisture resistance	1004		10
Hermetic seal	1014	Test condition A_1 for fine leaks (leak testing 2 hours after pressurization is acceptable); test condition C_1 for gross leaks, except that leak indicator fluid shall be maintained at +100°C ±5°C	10
External visual	2009		10
End points: Subgroup 1 Group A			
Subgroup 2			
Shock	2002	Non-operating; 500 G; t = 1 ms; 5 blows in each orientation	10
Constant acceleration	2001	Non-operating; 5000 G; in each orientation	10
End points: Subgroup 1 Group A			
Subgroup 3			
Salt atmosphere (corrosion)	1009		15
Subgroup 4 1/			
Steady-state operation life	1005	T = +85°C I_D = 150 mA, DC 1000 hours	7
End points: Subgroup 1 Group A			

See footnote at end of table.

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TABLE V. Group C inspection - Continued.

Inspection	MIL-STD-883 method	Details	LTPD
Subgroup 5			
Nuclear survivability	1019 1017		15
End points:			
Subgroup 1 Group A	-		

1/ Failed devices from the same inspection lot may be used for all subgroups when electrical or optical end-point measurements are not required.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - AV
Navy - AS, EC
Air Force - 17, 19, 70, 71, 80, 84, 90, 99
DLA - ES
NASA - NA

User activities:
Air Force - 13, 14

Preparing activity:
Army - CR

Agent:
DLA - ES

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